Checkpoint C | Version 3 | Solutions

*Each skill earns **Pass** or **Not Yet**. Some questions contain multiple skills. Grade each skill on its own when it's part of a full question. A **Pass** is given for work showing the core skill. A **Not Yet** is given when there's not sufficient evidence of the core skill. Evidence anywhere on the submission is acceptable but evidence in the question area will be used for the initial grading. Students may request a regrade if evidence exists outside the original grading area and was not included in the original grading. Do not infer a method that the work does not support.*

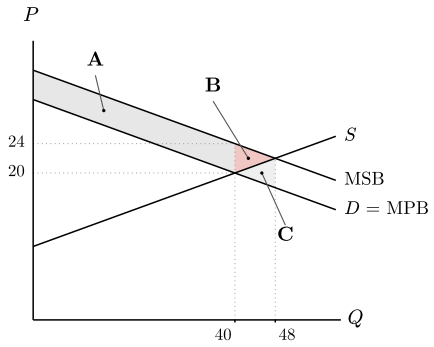
Lovely Lighting from Lumos Lanterns

Many Hogsmeade homeowners hang Lumos Lanterns outside their doors. In addition to providing lighting for the homeowner, each lantern also lights the lane for the neighbors, an external benefit of 4 coins per lantern. Prices are in coins and quantities in lanterns. The demand and supply equations are:

$$D: P_b = 30 - \frac{1}{4}Q_d \qquad S: P_s = 10 + \frac{1}{4}Q_s$$

C1.2 | Market Failure

The graph shows the market for lanterns: marginal private benefit (MPB), marginal social benefit (MSB), and supply (S). The lettered regions are for the questions.



Market Quantity ($\hat{Q}$): _____

Efficient Quantity (Q^*): _____

Deadweight Loss (*letter*): _____

Deadweight Loss (*area*): _____

For the Last Lantern Traded: MSB: _____ / MC: _____

The Market Trades: ☐ too many ☐ too few lanterns

Answers: market quantity **40 lanterns**; efficient quantity **48 lanterns**; DWL region **B**; DWL area **16 coins**; last lantern traded: **MSB = 24**, **MC = 20**; **too few** lanterns.

The 4-coin external benefit places MSB above MPB. Market equilibrium uses private benefit; efficiency counts the neighbors' benefit:

$$\text{MSB} = 30 - \frac{Q}{4} + 4 = 34 - \frac{Q}{4}$$
$$30 - \frac{Q}{4} = 10 + \frac{Q}{4} \Rightarrow \hat{Q} = 40, \quad \hat{P} = 20$$
$$34 - \frac{Q}{4} = 10 + \frac{Q}{4} \Rightarrow Q^* = 48.$$

At the last lantern traded, $\text{MSB} = 34 - 40/4 = 24$ and $\text{MC} = 10 + 40/4 = 20$. Trades between 40 and 48 would benefit society more than they cost, but private buyers do not purchase them. Their lost surplus is region B:

$$\text{DWL} = \frac{1}{2} \times 4 \times (48 - 40) = 16 \text{ coins.}$$

A is the external benefit of lanterns already traded. C is the gap between MC and MPB on the additional socially worthwhile trades.

Pass (C1.2): Read the market and efficient quantities from the correct intersections, identify B and calculate its area, and compare MSB and MC at the market quantity to identify too few trades. No separate MSB-gap answer is required.

Checkpoint C | Version 3 | Solutions

C2.2 | A Subsidy for Lanterns

To brighten the village, the Ministry pays sellers a subsidy of 6 coins per Lumos Lantern sold. For this question, ignore the external benefit: demand is the marginal benefit (MB) of a lantern.

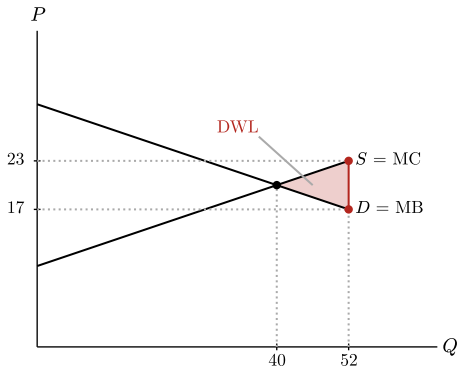
Buyer's Price (P_b): _____

Seller's Price (P_s): _____

Subsidy Quantity (Q^{Sub}): _____

Deadweight Loss (DWL): _____

Last Lantern Traded, MC vs. MB: ☐ above ☐ below



Answers: buyer's price **17 coins**; seller's price **23 coins**; subsidy quantity **52 lanterns**; DWL **36 coins**; MC is **above** MB for the last lantern traded.

Ignoring the external benefit, the unsubsidized market is efficient at $Q = 40$ and $P = 20$. The subsidy makes the seller's price 6 coins higher than the buyer's:

$$P_s = P_b + 6 \Rightarrow 10 + \frac{Q}{4} = 30 - \frac{Q}{4} + 6 \Rightarrow Q^{Sub} = 52.$$

$$P_b = 30 - \frac{52}{4} = 17, \qquad P_s = 10 + \frac{52}{4} = 23.$$

Check: $23 - 17 = 6$. At the last trade, MC is 23 and MB is 17. The added trades between 40 and 52 cost more than their private benefit, giving:

$$\text{DWL} = \frac{1}{2} \times 6 \times (52 - 40) = 36 \text{ coins}.$$

Pass (C2.2): Use the subsidy wedge to find both prices and quantity, identify MC above MB, and calculate DWL using the no-externality baseline of 40, rather than the 48 from C1.2.

Checkpoint C | Version 3 | Solutions

Cauldrons For Days

The Standard Cauldron is a magical tool used widely in the making of various potions with many buyers and many sellers. The demand and supply curves for the Standard Cauldron can be represented by the following relationships, with prices in coins, and quantities in cauldrons.

$$D: P_b = 110 - \frac{1}{2}Q_d \qquad S: P_s = 20 + Q_s$$

The metal for each cauldron is smelted in foundries outside Hogsmeade, and the fumes drift over the village. The fumes cost residents 9 coins per cauldron made, a cost neither buyers nor sellers pay.

C1.1 | Market Failure

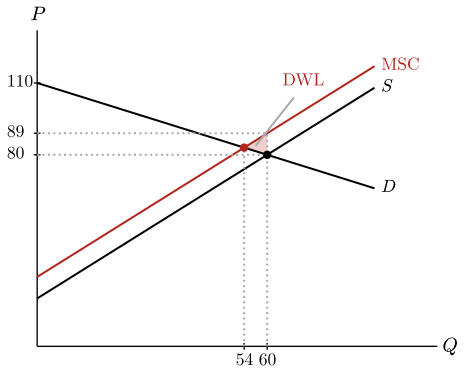
Start by using a graph and algebra to describe the market failure. Graph demand, supply, and MSC, and shade the deadweight loss.

Market Quantity ($\hat{Q}$): _____

Efficient Quantity (Q^*): _____

Deadweight Loss (DWL): _____

The Market Trades: ☐ too much ☐ too little



Answers: market quantity **60 cauldrons**; efficient quantity **54 cauldrons**; DWL **27 coins**; **too much**. Draw MSC $= 29 + Q$ and shade the triangle between MSC and demand from 54 to 60.

Add the 9-coin external cost to supply. The private and social marginal conditions give:

$$\text{MSC} = 20 + Q + 9 = 29 + Q$$
$$110 - \frac{1}{2}Q = 20 + Q \Rightarrow \hat{Q} = 60, \quad \hat{P} = 80$$
$$110 - \frac{1}{2}Q = 29 + Q \Rightarrow Q^* = 54.$$

At $Q = 60$, MSC is 89 while MB is 80. Trades beyond 54 cost society more than they benefit buyers. The DWL triangle has corners $(54, 83)$, $(60, 89)$, and $(60, 80)$:

$$\text{DWL} = \frac{1}{2} \times 9 \times (60 - 54) = 27 \text{ coins.}$$

Pass (C1.1): Construct MSC, calculate market and efficient quantities, correctly shade and calculate DWL, and identify overproduction. MSC is required on the graph, not as a separate equation blank.

Checkpoint C | Version 3 | Solutions

C2.1 | A Tax for Revenue

Needing revenue, and not yet knowing the size of the externality, the Ministry imposes a tax of 12 coins per cauldron. For this question, ignore the external cost.

Buyer's Price (P_b): _____

Seller's Price (P_s): _____

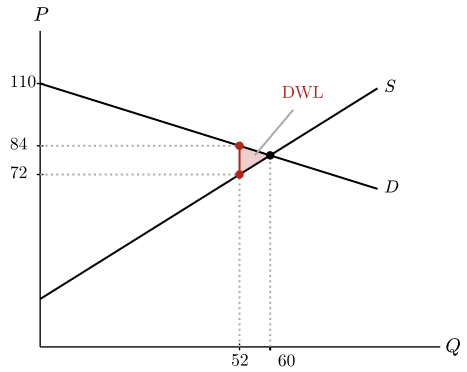
Post-Tax Quantity (Q^{Tax}): _____

Tax Incidence on Buyers: _____

Tax Incidence on Sellers: _____

Deadweight Loss (DWL): _____

Who Bears More of the Tax? ☐ buyers ☐ sellers ☐ equal



Answers: buyer's price **84 coins**; seller's price **72 coins**; post-tax quantity **52 cauldrons**; incidence **4 coins per cauldron on buyers** and **8 on sellers**; DWL **48 coins**; **sellers** bear more.

The tax separates the two prices by 12 coins:

$$P_b = P_s + 12 \Rightarrow 110 - \frac{1}{2}Q = 20 + Q + 12 \Rightarrow Q^{Tax} = 52.$$

Substituting into demand and supply gives $P_b = 84$ and $P_s = 72$. Check: $84 - 72 = 12$. Relative to the untaxed price of 80, buyers pay $84 - 80 = 4$ more and sellers receive $80 - 72 = 8$ less. Thus sellers bear more.

Ignoring the external cost, the efficient quantity is 60. The tax prevents 8 mutually beneficial trades:

$$\text{DWL} = \frac{1}{2} \times 12 \times (60 - 52) = 48 \text{ coins.}$$

Pass (C2.1): Use the tax wedge to find both prices and quantity, calculate each burden against the untaxed price, identify sellers as bearing more, and calculate DWL with the external cost ignored.

Checkpoint C | Version 3 | Solutions

C3.1 | Designing the Policy

The Ministry of Magic has hired you to design a policy that corrects the market failure caused by the negative externality of cauldron production. Propose the policy you recommend and how it would impact the market for cauldrons and their users.

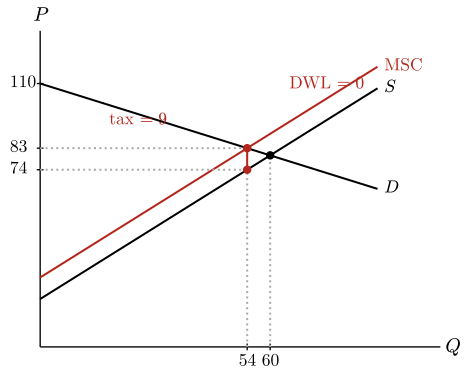
Policy Type: _____

Policy Size: _____

Post-Policy Quantity (Q^{τ}): _____

Post-Policy Deadweight Loss: _____

Post-Policy Quantity vs. Efficient: ☐ above ☐ at ☐ below



Answers: a per-unit **tax** of **9 coins per cauldron**; post-policy quantity **54 cauldrons**; remaining DWL **0 coins**; quantity is **at** the efficient level.

Count the external cost again and replace the earlier 12-coin revenue tax with a tax equal to the 9-coin external cost. Supply plus the tax then coincides with MSC:

$$P_b = P_s + 9 \Rightarrow 110 - \frac{1}{2}Q = 20 + Q + 9 \Rightarrow Q^{\tau} = 54.$$

Compared with the unregulated market, the corrective tax removes the socially costly trades between 54 and 60. At 54, MB equals MSC, so the policy reaches efficiency and leaves no deadweight loss.

Users pay 83 coins and sellers keep 74 coins. Relative to the unregulated market, users buy 6 fewer cauldrons and residents experience less harm from the fumes. These policy prices explain the effect; they are not extra required answer blanks.

Pass (C3.1): Choose a tax of 9, calculate its quantity, identify that quantity as efficient, and give zero remaining DWL. Compare the corrective policy with efficiency, not the earlier revenue tax. Assess this skill independently of arithmetic slips in other skills.